STM32F411-BASED ELECTRONIC LOCK

Product Scope & System Requirements

Normative V1 definition for the functional MVP and the staged power-management increment

Document purpose

Define the product boundary, externally observable behavior, safety intent, measurable requirements, verification methods and release gates. The root README separately defines the software architecture used to satisfy this specification.

Field	Value
Project	STM32F411-Based Electronic Lock
Product baseline	V1 functional engineering prototype
Document version	1.0
Status	CANDIDATE BASELINE - OWNER APPROVAL REQUIRED
Owner	Joao Pedro Rey
Target	STM32F411CEU6 / STM32CubeIDE / FreeRTOS
Date	2026-08-12

Controlled engineering document - approval converts this candidate into the normative V1 baseline.

Document Control and Approval

This document replaces the product-scope and requirement authority formerly embedded in the historical Architecture-V1 reference. It does not erase that file; the old document remains historical evidence until a deliberate repository migration is approved.

Role	Name	Decision	Date
Document owner	Joao Pedro Rey	Pending approval	-
Technical reviewer	Project review process	Candidate prepared	2026-08-12
Implementation baseline	Repository main branch	Apply after approval	-

Revision History

Version	Date	Status	Summary
1.0	2026-08-12	Candidate	First consolidated STM32F411/FreeRTOS product specification; replaces obsolete STM32F103/superloop scope assumptions.

Normative Authority

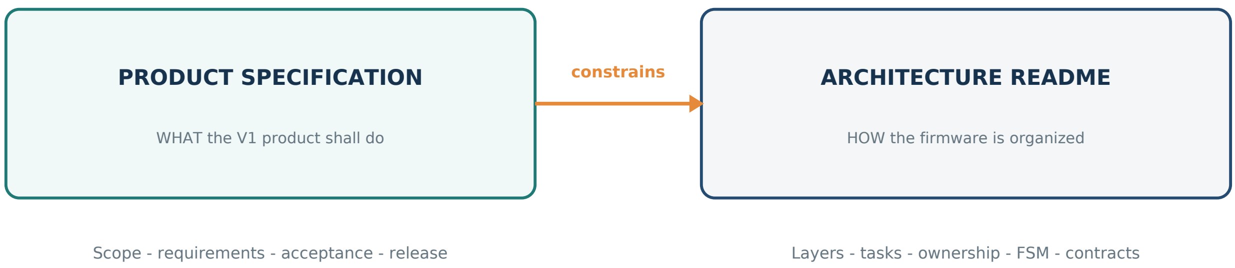

Figure 1 - Normative separation between product specification and software architecture.

Source	Authority
This product specification	Product scope, system behavior, requirement IDs, verification and release acceptance.
Root README.md	Software layers, task model, ownership, event model, state-machine implementation and documentation standard.
Electronic-Lock.ioc	Intended STM32 pin, clock and peripheral configuration.
Public module headers	Exact symbol-level API contracts, within the product and architecture constraints.
Historical references	Context and traceability only; no authority over this baseline.

Conflict rule

A discovered conflict is resolved by updating the higher-authority normative source before or together with affected implementation. Code behavior does not silently redefine a requirement.

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1. Purpose and Normative Use

The V1 product shall be a deterministic and testable electronic lock engineering prototype. It receives a four-digit PIN from a matrix keyboard, validates that candidate, commands one actuator for a bounded interval and communicates product state through a character display, status LEDs and a passive buzzer.

> Normative wording
>
> Shall means mandatory. Should means recommended unless a documented deviation is approved. May means permitted. Statements without these terms provide context and do not replace an identified requirement.

1.1 Intended Audience

- The project owner approving product scope and release acceptance.
- Firmware engineers implementing services, state machines, drivers and FreeRTOS integration.
- Hardware engineers selecting the actuator, power stage, supply and battery path.
- Reviewers deriving tests, reviewing changes and qualifying the V1 release.

1.2 Product Claim

V1 is a functional and reliable engineering prototype. It is not a certified lock, a production credential system or a claim of physical-security resistance. The firmware knows the electrical command issued to the actuator; without a position sensor it does not know the mechanical state of the door or latch.

2. Product Definition

2.1 Product Objective

Deliver a one-month functional MVP that demonstrates a safe boot, reliable credential entry, exact authentication, bounded unlock, automatic relock, lockout after repeated failures, non-blocking feedback and controlled fault behavior on STM32F411CEU6.

2.2 Success Measures

- No observed unintended actuator energization during the approved boot/reset test matrix.
- No invalid, incomplete, ambiguous or overflow-corrupted input can authorize unlock.
- Every authorized unlock ends at its bounded deadline even if the user interface fails.
- Every product state and transition has objective host or target evidence.
- The release build completes the 24-hour repeated-operation test without deadlock or unintended unlock.
- The architecture remains understandable: two project tasks, one input queue and one authoritative product FSM.

2.3 Engineering Principles

Principle	Consequence
Safety before interface	Actuator-safe initialization and deadline processing take precedence over display and feedback.
Non-blocking product behavior	Human-scale durations are state plus timestamps, never delays that stop the product.
Explicit ownership	Every resource and mutable state has one normal execution owner.
Bounded failure	Every hardware wait, retry and queue action has a finite outcome.
Measured complexity	New tasks, queues or asynchronous mechanisms require measured need.
Traceability	Requirement IDs connect design, implementation, tests and release evidence.

3. Scope Baseline

3.1 Functional MVP - Included

- STM32F411CEU6 controller using the committed STM32CubeIDE/CubeMX project.

- 4x4 matrix keyboard connected directly to MCU GPIOs.
- HD44780-compatible 16x2 LCD in 4-bit mode through PCF8574 on I2C.
- Fixed four-digit development credential compiled into firmware.
- Confirmation with # and clear/cancel behavior with *.
- Status LEDs and passive PWM buzzer with non-blocking patterns.
- One lock actuator through a protected power stage with safe polarity and maximum-on deadline.
- Entry timeout, failed-attempt counter, temporary lockout and full product fault state.
- FreeRTOS execution with two project-owned tasks and one keyboard event queue.
- Static memory, host tests for pure logic and target acceptance tests.
- Power Management service boundary implementing PM0 and preserving the staged low-power path.

3.2 V1 Power Increment - Included After Functional MVP

- UI inactivity policy and nonessential-output shutdown.
- Tickless idle after measured timing qualification.
- STOP mode with validated keyboard wake and time reconciliation.
- Battery measurement and low/critical policy after hardware thresholds are approved.

3.3 Explicitly Excluded

- User-changeable or multiple credentials, roles, administrators or audit log.
- Persistent credential database in Flash, EEPROM, secure element or remote service.
- Bluetooth, Wi-Fi, RFID, NFC, biometrics, mobile application or cloud backend.
- Cryptographic credential storage, tamper detection or resistance to firmware extraction.
- Mechanical door/latch sensing or closed-loop confirmation of lock position.
- OTA, custom bootloader, remote update or remote unlock.
- Physical-security, functional-safety or regulatory certification.
- A universal lock framework or speculative support for unrelated products.

> Scope guardrail
>
> A feature outside the functional MVP cannot displace an actuator-safety, integration, verification or release item. PM2-PM4 do not block the one-month functional MVP unless explicitly promoted by the owner.

4. System Context and Stakeholders

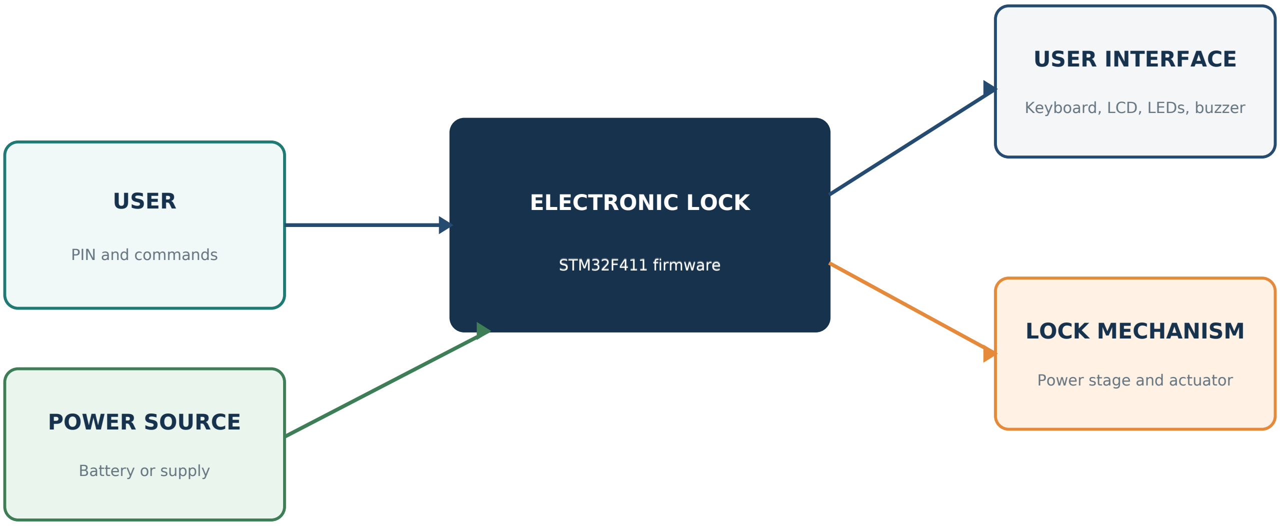

Figure 2 - Physical product boundary and external interactions.

Actor or subsystem	Interaction	Boundary
User	Enters PIN, confirms, cancels and observes feedback.	Cannot configure credentials or product policy in V1.
Technician/developer	Programs, diagnoses and validates the target.	Debug interface is not an end-user interface.
Power source	Supplies MCU, UI and actuator.	Power disturbance must lead to a safe command.
Lock mechanism	Converts the electrical command into mechanical action.	No mechanical feedback is available in V1.
Firmware	Applies access policy and drives configured interfaces.	Does not claim physical position or tamper resistance.

5. Operational Concept and Use Cases

5.1 UC-01 Authorized Access

1. The product starts with the actuator safe and completes critical initialization.
2. The locked prompt and locked indication are presented.
3. The user enters four decimal digits; the display shows only masking markers.
4. The user confirms with #.
5. Authentication compares the complete candidate with the configured V1 credential.
6. On success, the product clears the candidate, resets failure count and commands a three-second unlock.
7. Success feedback progresses while the unlock deadline remains independently active.
8. At deadline expiry, the actuator returns safe and the product returns to locked idle.

5.2 Alternate and Failure Use Cases

Use case	Trigger	Expected outcome
UC-02 Invalid credential	Complete candidate does not match.	Erase candidate, increment failure count, deny access, remain safe.
UC-03 Incomplete confirmation	# pressed before four digits.	Do not authenticate; keep safe; provide bounded incomplete feedback.
UC-04 Cancel	* pressed.	Clear non-empty candidate; with empty candidate return to locked idle.
UC-05 Entry timeout	15 seconds of entry inactivity.	Erase candidate and return to locked idle.
UC-06 Lockout	Third consecutive invalid credential.	Reject authentication for 30 seconds; remain safe.
UC-07 Degraded UI	LCD, LED, buzzer or backlight fails.	Preserve access policy and deadlines; degrade or fault by classification.
UC-08 Critical failure	Time, actuator safety, input integrity or RTOS integrity fails.	Force safe and enter FAULT.
UC-09 Reset during unlock	Reset/brownout/watchdog while energized.	Hardware and startup path restore safe state immediately.
UC-10 Power transition	Approved inactivity and sleep guards.	Reduce power without weakening product behavior.

6. Product Behavior and States

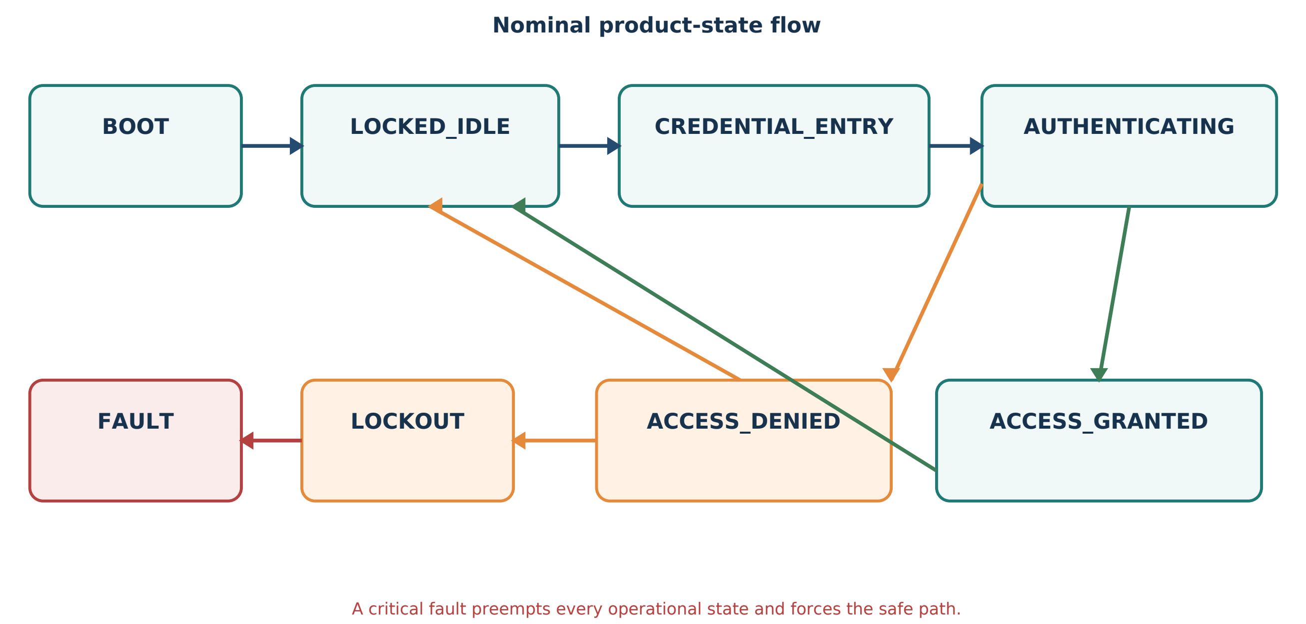

Figure 3 - Normative product states. Detailed transition ownership is defined in the root README.

State	Externally observable purpose	Actuator
BOOT	Establish safe output and decide whether critical startup succeeded.	Safe
LOCKED_IDLE	Present locked prompt and wait for a new credential.	Safe
CREDENTIAL_ENTRY	Collect, mask, clear and time the candidate.	Safe
AUTHENTICATING	Perform exact side-effect-free validation.	Safe
ACCESS_GRANTED	Maintain one bounded authorized unlock and success feedback.	Energized only within deadline
ACCESS_DENIED	Present bounded denial and choose retry or lockout.	Safe
LOCKOUT	Reject authentication until the lockout deadline expires.	Safe
FAULT	Latch critical failure, reject unlock and expose limited fault indication.	Safe

State invariant

The actuator is electrically safe in every state except the bounded energized portion of ACCESS_GRANTED. A critical fault preempts normal user-interface events.

7. Configuration Baseline

Parameter	Baseline	Ownership and rule
Credential length	4 digits	Fixed V1 product policy.
Credential value	Private build configuration	Never published or logged.
Entry timeout	15 s	Restarted after each accepted entry action.
Unlock duration	3 s	Subject to measured actuator thermal limit.
Failed attempts	3	Consecutive and saturating.
Lockout	30 s	No credential authentication while active.
Keyboard period	1 ms	Stable periodic task acquisition.
Debounce	40 ms	Tune only after target measurement.
Application heartbeat	10 ms maximum	Deadline and effect update ceiling.
I2C transaction timeout	20 ms maximum	Finite failure ceiling, not normal duration.
Keyboard event queue	8 entries	Measure high-water mark under stress.

Verification method key

Inspection: document/code review. Analysis: calculation or static reasoning. Test: objective execution with pass/fail evidence. Demonstration: observed end-to-end behavior on target.

8. Functional Requirements

FR requirements define user-visible and product-state behavior.

ID	Normative requirement	Release	Verification
FR-001	On power-on, reset or watchdog recovery, the system shall establish the actuator electrical safe state before normal application behavior is enabled.	**MVP**	Test + demonstration
FR-002	The system shall initialize all configured subsystems in a deterministic sequence and retain the result of each initialization step.	**MVP**	Inspection + test
FR-003	The system shall enter normal locked operation only when every dependency classified as critical has initialized successfully.	**MVP**	Fault-injection test
FR-004	The matrix keyboard shall produce one logical click action for one valid physical press-and-release cycle.	**MVP**	Target test
FR-005	Electrically ambiguous simultaneous-key or ghosting conditions shall not be accepted as credential digits.	**MVP**	Target test
FR-006	A digit pressed in locked idle shall start a credential-entry session and become the first candidate digit.	**MVP**	Host + target test
FR-007	The credential-entry function shall accept only decimal digits and shall store no more than four digits.	**MVP**	Unit test
FR-008	The user interface shall display only a masked indication of entered credential length and shall never render raw credential digits.	**MVP**	Inspection + test
FR-009	Confirmation shall request authentication only when the candidate contains exactly four digits.	**MVP**	Unit test
FR-010	Confirmation of an incomplete candidate shall keep the actuator safe, provide bounded feedback and preserve a valid entry session.	**MVP**	Unit + product test
FR-011	Cancel pressed with a non-empty candidate shall erase the candidate; cancel pressed with an empty candidate shall terminate the entry session.	**MVP**	Unit test
FR-012	Credential-entry inactivity for 15 seconds shall erase the candidate and return the product to locked idle.	**MVP**	Unit + product test
FR-013	Authentication shall accept only an exact four-digit match to the configured V1 credential and shall have no hardware side effects.	**MVP**	Unit test
FR-014	Successful authentication shall reset the consecutive-failure counter before the next authentication session.	**MVP**	State-machine test
FR-015	Failed authentication shall erase the candidate and increment a saturating consecutive-failure counter.	**MVP**	State-machine test
FR-016	After three consecutive authentication failures, the product shall enter a 30-second lockout and reject credential authentication during that interval.	**MVP**	Product test
FR-017	Lockout completion shall reset the consecutive-failure counter and return the product to locked idle.	**MVP**	State-machine test
FR-018	Successful authentication shall request a finite actuator unlock interval of 3 seconds.	**MVP**	Target measurement
FR-019	At unlock expiry, the actuator shall be commanded safe independently of display, LED or buzzer state or failure.	**MVP**	Fault-injection test
FR-020	A critical fault during authorized unlock shall immediately cancel the unlock request and invoke the actuator safe path.	**MVP**	Fault-injection test
FR-021	The product shall provide distinguishable visual states for boot, locked, entry, granted, denied, lockout and fault.	**MVP**	Demonstration
FR-022	The product shall provide bounded audible feedback for a valid key action, access granted, access denied and lockout.	**MVP**	Demonstration + timing test
FR-023	Visual and audible effects shall progress without blocking credential, fault or actuator-deadline processing.	**MVP**	Analysis + stress test
FR-024	The display function shall suppress retransmission when the requested logical view has not changed.	**MVP**	Unit + bus trace test

ID	Normative requirement	Release	Verification
FR-025	A recoverable user-interface failure shall be reported to the product controller without modifying authentication or actuator deadlines.	**MVP**	Fault-injection test
FR-026	A critical functional failure shall erase candidate data, force the actuator safe, reject unlock requests and enter the FAULT state.	**MVP**	Fault-injection test
FR-027	The product shall distinguish commanded actuator state from mechanically confirmed lock state and shall not report mechanical confirmation in V1.	**MVP**	Inspection + demonstration
FR-028	If keyboard-event ordering is lost because the event transport overflows, the product shall erase the candidate and require a new entry attempt.	**MVP**	Synthetic overflow test

8.1 Functional Behavior Decision Table

Condition	Permitted outcome	Prohibited outcome
Locked idle + digit	Begin entry and retain the digit as position one.	Dropping the first digit or authenticating immediately.
Entry below four digits + #	Report incomplete and remain safe.	Calling Authentication or energizing the actuator.
Entry at four digits + extra digit	Reject/ignore without modifying memory bounds.	Writing a fifth digit or shifting the candidate.
Entry at four digits + #	Authenticate exactly once and erase candidate data.	Repeated authentication from a retained key event.
Third consecutive failure	Complete bounded denial, then enter lockout.	Returning to normal credential entry before lockout.
Unlock deadline or critical fault	Invoke the safe actuator path immediately.	Waiting for display, LED or sound completion.
Input queue overflow	Invalidate candidate and request a new attempt.	Authenticating an event stream with lost ordering.

Decision-table precedence

Critical fault and actuator-deadline handling have higher priority than normal input and feedback events. This precedence applies even if a lower-priority event was received first in the same Application Task cycle.

9. Safety Requirements

Safety requirements protect the actuator-safe state and bound hazardous energization. They do not claim certification.

ID	Normative requirement	Release	Verification
SAF-001	The actuator safe electrical level shall be defined by board configuration and verified against the selected power stage before actuator-driver integration.	**MVP**	Inspection + measurement
SAF-002	The actuator power stage shall default to the safe state while the MCU output is reset, high-impedance or not yet configured.	**MVP**	Schematic inspection + test
SAF-003	Every public unlock command shall require a finite nonzero duration; an unlimited public unlock operation is prohibited.	**MVP**	API inspection + unit test
SAF-004	The lock-actuator component shall enforce a redundant maximum-on deadline independent of normal user-interface behavior.	**MVP**	Unit + target test
SAF-005	Reset, brownout or watchdog recovery during actuator energization shall return the output to the safe electrical state without waiting for application initialization.	**MVP**	Target fault test
SAF-006	No LCD, LED, buzzer, keyboard or I2C failure shall extend actuator energization beyond its approved maximum interval.	**MVP**	Fault-injection test
SAF-007	The power stage shall include load-appropriate suppression and protection, including flyback protection when the actuator is inductive.	**MVP**	Schematic + hardware inspection
SAF-008	Actuator current, inrush, thermal limit and maximum continuous energization shall be documented before the final unlock interval is approved.	**MVP**	Analysis + measurement
SAF-009	Stack overflow, RTOS assertion and unrecoverable application integrity hooks shall invoke the shortest valid actuator safe path.	**MVP**	Code inspection + fault test
SAF-010	The FAULT state shall reject all credential and unlock requests until a controlled reset or separately approved recovery policy occurs.	**MVP**	State-machine test
SAF-011	Firmware shall not claim certified physical security, functional safety or confirmed door position for V1.	**MVP**	Documentation inspection
SAF-012	All safety-relevant target tests shall record firmware revision, board revision, build profile and test conditions.	**MVP**	Test-report inspection

10. Security and Credential Requirements

V1 security requirements prevent accidental credential exposure and define explicit prototype limitations.

ID	Normative requirement	Release	Verification
SEC-001	The configured development credential shall not appear in public documentation, display content or normal diagnostic output.	**MVP**	Repository + runtime inspection
SEC-002	The candidate credential shall use fixed-capacity storage with explicit length and bounds validation.	**MVP**	Code inspection + unit test
SEC-003	Candidate storage shall be erased after success, failure, cancellation, entry timeout, lockout, critical fault and reset initialization.	**MVP**	Unit + state-machine test
SEC-004	Authentication shall reject every candidate whose length differs from four without reading outside the supplied buffer.	**MVP**	Boundary test + analysis
SEC-005	The failure counter shall saturate at its configured maximum and shall not wrap to a lower value.	**MVP**	Unit test
SEC-006	During lockout, keyboard acquisition may continue for responsiveness but no candidate shall be authenticated.	**MVP**	Product test
SEC-007	A credential used for demonstration shall not be represented as suitable for a deployed security product.	**MVP**	Documentation inspection
SEC-008	Persistent credential storage, cryptographic protection and anti-tamper behavior are outside V1 and shall not be implied by the implementation.	**MVP**	Scope inspection
SEC-009	Replacement of the V1 authentication mechanism shall remain isolated behind the Authentication service contract.	**MVP**	Architecture inspection

11. Performance and Timing Requirements

Timing requirements define initial budgets and the evidence required to approve them.

ID	Normative requirement	Release	Verification
TIM-001	The FreeRTOS application tick shall initially be configured to 1 millisecond.	**MVP**	Configuration inspection
TIM-002	The keyboard acquisition task shall execute with a nominal 1-millisecond period using delay-until semantics.	**MVP**	Trace measurement
TIM-003	The initial keyboard debounce interval shall be 40 milliseconds and shall be tunable only from centralized configuration after target measurement.	**MVP**	Configuration + target test
TIM-004	The Application Task shall update product deadlines and non-blocking services at least once every 10 milliseconds under nominal operation.	**MVP**	Trace measurement
TIM-005	One synchronous I2C transaction shall use a finite timeout no greater than 20 milliseconds in the initial baseline.	**MVP**	Code inspection + fault test
TIM-006	Human-scale durations shall be represented by state and timestamps rather than blocking delays.	**MVP**	Static inspection + test
TIM-007	Elapsed-time calculations shall remain correct across unsigned 32-bit time-source rollover.	**MVP**	Host boundary test
TIM-008	Keyboard acquisition shall remain schedulable while the Application Task performs a bounded synchronous display transaction.	**MVP**	Timing analysis + target trace
TIM-009	The measured actuator safe-command latency after unlock expiry shall remain within an approved tolerance recorded in the target test report.	**MVP**	Target measurement
TIM-010	LED and sound update resolution shall be no coarser than the 10-millisecond Application Task heartbeat unless a pattern explicitly tolerates a lower rate.	**MVP**	Unit + target test
TIM-011	No retry loop shall be unbounded; retry count and worst-case elapsed time shall be documented.	**MVP**	Inspection + fault test
TIM-012	Worst-case task execution time, event latency, stack high-water mark and queue high-water mark shall be measured before release.	**MVP**	Target measurement

12. Reliability and Fault Requirements

Reliability requirements define finite failures, degradation boundaries, static memory and release endurance.

ID	Normative requirement	Release	Verification
REL-001	All public operations that can fail shall return a status that distinguishes successful completion from invalid input and dependency failure.	**MVP**	API + test inspection
REL-002	A module shall not return success when its contract could not be completed.	**MVP**	Fault-injection test
REL-003	Failed component operations shall preserve the last confirmed software state or mark that state unknown; they shall not silently commit an unperformed hardware change.	**MVP**	Unit + fault test
REL-004	Critical and degradable initialization failures shall be classified by the product controller according to the approved dependency table.	**MVP**	Initialization matrix test
REL-005	The LCD, status LEDs, buzzer and backlight shall be degradable only while authentication, time and actuator safety remain valid.	**MVP**	Fault-injection test
REL-006	The time source, actuator safe path, keyboard input path and RTOS execution objects shall be treated as critical to normal operation.	**MVP**	Inspection + fault test
REL-007	The firmware shall tolerate an absent or nonresponsive PCF8574 without deadlock or unbounded startup time.	**MVP**	Target disconnection test
REL-008	The keyboard event queue shall use static storage and a defined capacity of eight logical events in the initial baseline.	**MVP**	Configuration inspection
REL-009	Application-owned task stacks, control blocks, queues and module instances shall use static storage.	**MVP**	Link map + code inspection
REL-010	The application shall not use malloc, free or application-owned dynamic allocation.	**MVP**	Static inspection
REL-011	The release candidate shall complete at least 24 hours of repeated operation without deadlock, stack overflow, queue corruption or unintended unlock.	**MVP**	Long-run test
REL-012	CubeMX regeneration shall not erase product logic, and generated changes shall be reviewed for pin, clock and peripheral consistency.	**MVP**	Regeneration test + diff inspection

13. Power Management Requirements

Power requirements are staged. MVP items preserve the architecture; V1-PM items are qualified after normal operation is stable.

ID	Normative requirement	Release	Verification
PWR-001	The functional MVP shall avoid busy polling when a task can block or delay deterministically.	**MVP**	Static inspection + CPU idle measurement
PWR-002	A Power Management service boundary shall exist without owning a dedicated V1 task.	**MVP**	Architecture + API inspection
PWR-003	The Power Management service shall never authenticate a credential or issue an unlock command.	**MVP**	Dependency inspection + unit test
PWR-004	The product shall support a configurable inactivity policy for dimming or disabling nonessential UI outputs while remaining locked.	**V1-PM**	Product test
PWR-005	Deep sleep shall be prohibited while the product is in ACCESS_GRANTED or while any actuator unlock deadline is active.	**V1-PM**	Guard unit test + target test
PWR-006	A deep-sleep transition shall verify actuator-safe state, empty event transport, inactive peripheral transactions, preservable deadlines and armed wake sources.	**V1-PM**	Guard test
PWR-007	A rejected sleep guard shall return the product to a responsive idle power state without becoming a product fault unless a safety contract was violated.	**V1-PM**	State-machine test
PWR-008	Keyboard wake shall be treated as a wake indication; a wake-causing transition shall pass normal scan and debounce before becoming a credential action.	**V1-PM**	Target test
PWR-009	Resume shall restore the system clock, required Platform peripherals and monotonic-time accounting before normal application events are accepted.	**V1-PM**	Target test
PWR-010	Tickless idle shall be enabled only after measured wake latency preserves keyboard and actuator timing requirements.	**V1-PM**	Measurement + analysis
PWR-011	STOP mode shall be enabled only after the exact wake pins, low-power time source and peripheral restore sequence are validated on the target board.	**V1-PM**	Target qualification
PWR-012	Battery thresholds and actuation policy shall not be implemented until battery chemistry, voltage range, sensing accuracy and actuator voltage limits are documented.	**V1-PM**	Hardware decision review
PWR-013	Battery-state filtering shall include documented hysteresis and shall represent unknown measurement as a distinct conservative state.	**V1-PM**	Unit + target test
PWR-014	Power optimization shall not weaken authentication, lockout, fault or actuator-safe behavior.	**V1-PM**	Regression test

14. Hardware and Electrical Requirements

Hardware requirements bind the current STM32F411 board and identify evidence required before safe release.

ID	Normative requirement	Release	Verification
HWR-001	The target controller shall be STM32F411CEU6 operating from the clock configuration committed in Electronic-Lock.ioc, initially 100 MHz.	**MVP**	Configuration + target inspection
HWR-002	The 4x4 matrix keyboard shall use PA0-PA3 as rows and PA4-PA7 as columns according to the committed board configuration.	**MVP**	Pin inspection + target test
HWR-003	The 16x2 HD44780-compatible display shall operate in 4-bit mode through a PCF8574 on I2C1 PB6/PB7.	**MVP**	Interface test
HWR-004	The real PCF8574 address and LCD backpack signal mapping shall be measured or inspected and centralized in board configuration before release.	**MVP**	Hardware inspection
HWR-005	TIM2 shall provide the microsecond timing source without conflicting with the FreeRTOS tick source.	**MVP**	Configuration + timing test
HWR-006	TIM3 channel 1 on PB4 shall drive the passive buzzer PWM path.	**MVP**	Frequency measurement
HWR-007	TIM4 channel 4 on PB9 shall drive the LCD backlight PWM path.	**MVP**	Duty/frequency measurement
HWR-008	PA12 and PA15 shall drive the red and blue status LED paths with explicitly configured active levels.	**MVP**	Polarity test
HWR-009	PB8 shall be reserved for the lock-actuator command and shall have an externally safe reset behavior.	**MVP**	Schematic + reset measurement
HWR-010	I2C pull-up values shall support the configured Fast Mode rise time and bus capacitance at the actual supply voltage.	**MVP**	Calculation + oscilloscope test
HWR-011	The actuator supply and PCB wiring shall prevent load switching from causing MCU brownout, reset or invalid logic levels.	**MVP**	Power integrity test
HWR-012	The hardware baseline shall identify board revision, actuator part number, power-stage revision and supply specification used for release tests.	**MVP**	Configuration audit

15. Software and Interface Constraints

These product constraints freeze only architecture choices required for predictable V1 delivery and verification.

ID	Normative requirement	Release	Verification
SWC-001	Product behavior shall be implemented by one explicit Lock Controller state machine with BOOT, LOCKED_IDLE, CREDENTIAL_ENTRY, AUTHENTICATING, ACCESS_GRANTED, ACCESS_DENIED, LOCKOUT and FAULT states.	**MVP**	Architecture + test inspection
SWC-002	The MVP shall contain exactly two project-owned runtime tasks: Keyboard Task and Application Task.	**MVP**	RTOS configuration inspection
SWC-003	The Keyboard Task shall own matrix acquisition; the Application Task shall own services, product state and output drivers after startup.	**MVP**	Ownership inspection
SWC-004	Keyboard-to-application communication shall use one bounded FIFO queue containing value-type logical key/action events.	**MVP**	Code inspection + queue test
SWC-005	Services and component drivers shall not include FreeRTOS or CMSIS-RTOS headers.	**MVP**	Dependency scan
SWC-006	Application and service public contracts shall not expose STM32 HAL types, pins, addresses or register details.	**MVP**	Header inspection
SWC-007	Dependency direction shall remain Application to Services to Components to Platform to STM32 HAL/CMSIS.	**MVP**	Include/link dependency scan
SWC-008	Core/Src/main.c shall contain only generated initialization, minimal product bootstrap and scheduler-start integration inside preserved user sections.	**MVP**	Code inspection + regeneration test
SWC-009	Each mutable state and hardware resource shall have one documented owner in normal execution.	**MVP**	Ownership review
SWC-010	No mutex is required by the V1 baseline; adding one shall require an approved architecture change with measured justification.	**MVP**	Architecture inspection
SWC-011	Component drivers shall expose device behavior, not credential, attempt, lockout or product-state policy.	**MVP**	API + dependency inspection
SWC-012	Adapters shall translate explicit component contracts and shall contain no product behavior.	**MVP**	Code inspection
SWC-013	Configuration shall separate product policy from board pins, addresses, channels and electrical polarity.	**MVP**	Configuration inspection
SWC-014	Public APIs shall document parameter direction, units, ownership, lifetime, preconditions, task context, blocking behavior and status results where applicable.	**MVP**	Documentation review
SWC-015	A third task, additional queue, DMA bus path or software timer shall require evidence that the two-task baseline cannot meet a measured requirement.	**MVP**	Architecture change review

16. Documentation and Configuration Requirements

Documentation is part of the release baseline, not a post-implementation activity.

ID	Normative requirement	Release	Verification
DOC-001	The root README shall remain the normative software-architecture baseline and shall identify this document as the normative product-scope and system-requirements baseline.	**MVP**	Repository inspection
DOC-002	Module public contracts and README files shall be updated in the same change that modifies public behavior.	**MVP**	Change review
DOC-003	All repository source documentation shall use English and shall match real filenames, symbols and units.	**MVP**	Documentation lint/review
DOC-004	Every source file, public type, structure field and public function shall follow the documentation template frozen in the root README.	**MVP**	Doxygen review
DOC-005	Each module shall document responsibilities, non-responsibilities, dependencies, timing, errors, constraints, tests and limitations.	**MVP**	README review
DOC-006	Architecture or product-requirement changes shall update the normative document before or together with implementation.	**MVP**	Change-control audit
DOC-007	Release evidence shall identify requirement and acceptance-test IDs rather than duplicating divergent prose.	**MVP**	Test-report review
DOC-008	The release tag shall contain the same product-specification and architecture revisions used to qualify the firmware.	**MVP**	Release audit

17. Power Management Staging

Power optimization follows measured increments so it cannot destabilize the functional access-control baseline.

Figure 4 - Staged V1 power roadmap.

Stage	Capability	Promotion evidence
PM0	No busy polling; deterministic task blocking/delay.	Functional MVP static review and CPU-idle measurement.
PM1	Backlight and feedback inactivity policy.	No lost first input; measurable current reduction.
PM2	FreeRTOS tickless idle.	Wake latency and task-period margin measured.
PM3	STM32 STOP mode with keyboard wake.	Clock/time/peripheral resume and all sleep guards validated.
PM4	Battery normal/low/critical/unknown policy.	Hardware thresholds, accuracy, hysteresis and actuation limits approved.

Power-state separation

Operational power state is orthogonal to the lock-product state. ACCESS_GRANTED is always active; LOCKED_IDLE may later be active or idle. Power Management can request resource changes but can never grant access.

18. Verification and Acceptance

18.1 Verification Levels

Level	Subjects	Evidence
Host unit	Authentication, credential entry, time calculations, product FSM, PM guards.	Repeatable native C test report.
Host component	Hardware-independent logic through fake Platform interfaces.	Boundary and failure-path results.
Target component	GPIO polarity, I2C, PWM, LCD, keyboard, time and actuator.	Instrumented target test report.
RTOS integration	Task cadence, event transport, ownership, stacks and CPU idle.	Trace and high-water measurements.
Product integration	All normal, denial, lockout, degraded and fault use cases.	Acceptance matrix with revision metadata.
Release	Long run, critical fault injection and build reproducibility.	Signed release checklist.

18.2 Acceptance Test Matrix

ID	Scenario	Procedure	Pass criterion	Requirements
AT-001	Safe boot	Perform 20 cold boots and 20 resets while observing PB8/power-stage command.	No unintended actuator energization; product reaches locked idle or explicit fault.	FR-001, FR-002, FR-003, SAF-001, SAF-002
AT-002	Authorized access	Enter the configured four-digit credential and confirm.	Access is granted once, success feedback starts and actuator returns safe after 3 seconds.	FR-006-FR-014, FR-018, FR-019
AT-003	Credential rejection	Test each wrong digit position, incomplete candidates and extra digits.	No invalid candidate energizes the actuator; bounds and feedback remain correct.	FR-007-FR-010, FR-013, SEC-002, SEC-004
AT-004	Cancel and timeout	Cancel at every candidate length and wait 15 seconds at every nonzero length.	Candidate is erased and product returns to the specified locked input state.	FR-011, FR-012, SEC-003
AT-005	Lockout	Submit three consecutive invalid credentials, then attempt input throughout 30 seconds.	No authentication occurs during lockout; safe output is maintained; normal input resumes after expiry.	FR-015-FR-017, SEC-005, SEC-006
AT-006	Debounce	Apply at least 100 deliberate presses including irregular press/release transitions.	One logical click per valid physical action and no duplicate credential digit.	FR-004, TIM-002, TIM-003
AT-007	Ambiguous keys	Apply supported simultaneous-key and ghosting patterns.	No ambiguous pattern becomes a credential digit.	FR-005
AT-008	Display disconnected	Disconnect PCF8574 before boot and during locked/access-granted operation.	No deadlock or unlock extension; degradation/fault follows policy.	FR-019, FR-025, REL-005, REL-007
AT-009	Reset during unlock	Reset and brownout the MCU at multiple points during actuator energization.	Power-stage command returns to safe state immediately according to measured hardware behavior.	SAF-002, SAF-005
AT-010	Critical fault during unlock	Inject time-source or actuator-path fault during authorized access.	Unlock is cancelled, safe path is invoked and FAULT rejects further input.	FR-020, FR-026, SAF-009, SAF-010
AT-011	Queue overflow	Synthetically saturate the keyboard event queue while a candidate is being entered.	Candidate is erased, ordering is not trusted and the user must retry.	FR-028, REL-008
AT-012	Rollover	Run entry, unlock, lockout and effect deadlines across 32-bit millisecond rollover.	All deadlines expire once at the intended elapsed duration.	TIM-007
AT-013	Timing and resources	Trace both tasks under worst supported UI and I2C behavior.	Periods, latency, stack margin, queue occupancy and CPU idle evidence satisfy the recorded budget.	TIM-001-TIM-012, REL-008, REL-009
AT-014	Long run	Execute repeated valid, invalid, cancel and timeout cycles for at least 24 hours.	No deadlock, corruption, stack overflow or unintended unlock occurs.	REL-011
AT-015	Power increment	Exercise every implemented PM state, guard rejection and wake source on target.	No sleep occurs during unlock; resume restores time/peripherals and wake input is debounced.	PWR-004-PWR-014

19. Traceability Summary

The acceptance matrix above provides direct test-to-requirement traceability. The following responsibility map prevents requirements from being orphaned during implementation.

Requirement family	Primary owner	Primary evidence
FR-001-FR-003	App Core and Lock Controller	Startup/fault matrix
FR-004-FR-012	Keyboard Task and Credential Entry	Host tests + keypad target tests
FR-013-FR-017	Authentication and Lock Controller	Host FSM/authentication suite
FR-018-FR-020	Lock Control and Lock Actuator	Instrumented actuator tests
FR-021-FR-025	Display, Status and Sound services	Fake-driver tests + demonstration
FR-026-FR-028	Lock Controller and App Core	Fault/overflow tests
SAF	Board, Lock Actuator and App failure hooks	Schematic, measurement and fault injection
SEC	Credential Entry and Authentication	Boundary tests + repository scan
TIM	App Core, Time Platform and services	Trace, rollover and WCET evidence
REL	All modules; integration owner	Failure tests, link map and long run

Requirement family	Primary owner	Primary evidence
PWR	Power Management service and board integration	Guard tests and current/wake measurements
HWR	Board configuration and component adapters	Inspection and instrumented target tests
SWC/DOC	Architecture and module owners	Static dependency and documentation review

No orphan rule

Every mandatory MVP requirement shall have an implementation owner and objective evidence before release. A requirement without evidence remains open even if the nominal feature appears to work.

20. Release Baseline and Change Control

20.1 Functional MVP Release Gates

- All requirements marked MVP are implemented or have a formally approved deviation.
- AT-001 through AT-014 pass on the identified board and release-equivalent build.
- No known defect can energize the actuator indefinitely or bypass exact authentication.
- Stack and queue margins, task timing, I2C timeout and actuator deadline evidence are recorded.
- The 24-hour repeated-operation test completes without a critical defect.
- README, module documentation, this specification and code correspond to the same commit/tag.
- All release-blocking hardware decisions are closed and reflected in board configuration.

20.2 V1 Power Increment Gates

- Only implemented V1-PM requirements are promoted into the release claim.
- AT-015 passes for every enabled low-power state and wake source.
- Current reduction, wake latency and time reconciliation are measured on the real board.
- Functional MVP acceptance tests regress successfully with power features enabled.

20.3 Requirement Change Procedure

1. State the observed problem, new product need or failed verification result.
2. Identify affected requirement, architecture and acceptance-test IDs.
3. Evaluate the smallest viable alternatives and safety/timing impact.
4. Update this specification before or with the implementation change.
5. Update architecture, module contracts, tests and release evidence in the same change.
6. Increment document version when approved behavior, scope or release acceptance changes.

21. Open Decisions and Release Risks

21.1 Controlled Open Decisions

ID	Decision required	Owner	Due / effect
OD-001	Exact actuator part, safe mechanical/electrical state, current, inrush and thermal limit.	Hardware	Before Lock Actuator API freeze; blocks MVP integration.
OD-002	Power-stage schematic, flyback/suppression and reset-state biasing.	Hardware	Before energized target testing; blocks MVP release.
OD-003	Actual PCF8574 address and backpack signal mapping.	Hardware/firmware	Before LCD integration release.
OD-004	Exact LED colors/active levels and visual/sound pattern table.	System/UX	Before service acceptance.
OD-005	LCD failure policy when remaining feedback channels are unavailable.	System	Before full FSM qualification.
OD-006	Host C test framework and CI execution command.	Firmware	Before service implementation completion.
OD-007	Watchdog inclusion in functional MVP or immediate post-MVP increment.	System	Before fault qualification.
OD-008	Battery chemistry, range, sensing circuit, measurement accuracy and actuator critical-voltage policy.	Hardware/system	Before PM4; does not block functional MVP.

21.2 Risk Register

ID	Risk	Impact	Mitigation
R-001	Actuator interface frozen before electrical selection.	Unsafe polarity, overcurrent or rework.	Close OD-001/OD-002 before final driver contract.
R-002	Load switching resets or corrupts MCU operation.	Uncontrolled product behavior.	Power integrity, suppression, decoupling and oscilloscope tests.
R-003	Incorrect LCD backpack mapping.	No display or invalid writes.	Measure mapping and centralize board configuration.
R-004	Key ghosting or bounce accepted as valid input.	Credential corruption.	Single-key policy, ambiguity rejection and repetitive target tests.
R-005	Missing I2C device consumes repeated maximum timeout.	Responsiveness and deadline risk.	Finite timeout, explicit degradation and bounded retry policy.
R-006	Premature tasks/interfaces delay MVP.	Schedule failure and synchronization defects.	Require measured need and architecture change review.
R-007	Credential leaks through repository or diagnostics.	Security exposure.	Private configuration, masking and repository/runtime scans.
R-008	Reset during unlock leaves power stage active.	Unsafe energization.	External safe bias, early GPIO safety and AT-009.
R-009	Deep sleep introduced before deadline/wake validation.	Missed input or unlock deadline.	Staged PM gates and full MVP regression.
R-010	Commanded state is mistaken for mechanical position.	False product claim.	Explicit limitation and future sensor requirement.

Appendix A - Glossary

Term	Definition
Candidate credential	The bounded four-digit value currently being entered; never the configured reference credential.
Commanded state	The electrical state requested by firmware; not confirmation of mechanical lock position.
Critical dependency	A dependency whose failure prevents normal operation because access or actuator safety cannot be trusted.
Degradable dependency	A feedback dependency whose loss may permit locked operation when all safety-critical paths remain valid.
Deadline	A rollover-safe elapsed-time condition that must cause a transition or safe action.
Functional MVP	The normal-operation V1 release target required before deep power optimization.
Lockout	A temporary state in which credential authentication is rejected after repeated failures.
Power stage	External circuitry that allows the MCU command to control actuator voltage/current safely.
V1-PM	Post-functional-MVP power-management increment governed by PWR requirements.

Appendix B - Review Checklist

- Does every statement that constrains release have a stable requirement ID?
- Is each requirement atomic, objectively verifiable and assigned to MVP, V1-PM or Deferred?
- Are actuator type, safe polarity and maximum energization based on hardware evidence?
- Can any display, sound, queue or input fault extend actuator energization?
- Can an incomplete, ambiguous, overlength or overflow-corrupted candidate reach authentication?
- Are all human-scale timings represented by rollover-safe state/deadline logic?
- Do all mandatory requirements have an owner and verification artifact?
- Does the root README architecture still satisfy this product specification without additional tasks or synchronization?
- Are power features staged behind measured wake, timing and regression evidence?
- Do repository documentation and release artifacts refer to the same version and tag?

End of controlled document